

INQUIRY INTO DATA CENTRES

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1. Executive summary

The Business Council of Australia (BCA) represents Australia's largest employers across all sectors, committed to building a stronger, more competitive, more productive Australia. We welcome the opportunity to respond to this inquiry on data centres in NSW.

Data centres are a critical part of the digital infrastructure that sit behind much of modern life. Parents can book medical appointments online. Teams can collaborate across cities. Students can access digital learning. Businesses can reach customers through cloud platforms and digital tools. All this is possible because of data centres processing millions of calculations every second. They support everyday services such as banking and transport apps, telehealth, education platforms and emergency systems, along with business-to-business systems. Data centres enable so much of our everyday lives and boost the productivity of Australia's businesses.

In simple terms, data centres keep the digital world running. Without them, everyday digital tools would slow down or disappear entirely.

While Australia needs data centres now, we also face a once in a generation opportunity. If we choose to be a data centre leader, Australia and NSW also gain:

- Increased GDP growth by capturing a larger share of the rapidly expanding global digital and AI economy
- Creation of high-value, long-term jobs across construction, engineering, operations and digital services
- Attraction of large-scale global investment, strengthening NSW and Australia's position as a regional technology hub
- Acceleration of renewable energy generation and storage, with data centre demand underpinning new clean energy projects at scale
- Improved economy-wide productivity through faster, more reliable digital services and AI adoption across all sectors
- Strengthened infrastructure and resilience, including grid upgrades, connectivity, and more robust delivery of critical services such as healthcare, finance and government systems

Data centres should be understood as critical infrastructure, and like traditional infrastructure, they underpin the productivity of the economy. Hospitals use them to access patient records and diagnostics. Small businesses rely on them for accounting systems and online stores. Governments rely on them to deliver services such as transport and welfare systems.

As with any major infrastructure or project, planning and regulatory requirements apply. Data centres require electricity, water, land and network capacity, so development should be carefully considered.

Industry is already addressing many of these challenges. Data centre operators are building renewable energy generation and storage, more efficient cooling technologies and infrastructure that strengthens water systems. Data centre investment helps finance new renewable energy generation and grid upgrades that benefit other users.

The global data centre market is growing quickly. Worldwide data centre capital expenditure may exceed US\$1 trillion each year by 2029 as AI and digital services expand.¹ Earlier phases of the industry focused on flexibility and choice. Today the focus is speed, with operators competing to deliver new capacity quickly.

And Australia is an increasingly attractive destination for digital infrastructure investment, supported by our renewable energy, strong connectivity, available land, stable political environment and a growing technology sector. Yet our position cannot be assumed to continue.

¹ Commonwealth Bank, *AI: Boom, Bubble or Both?*, February 2026, p.1

Competition across the Asia-Pacific region is intense. Singapore, Japan, Malaysia and other markets are also attracting investment, and NSW sits in the middle of this competition.

Governments need to recognise data centres as enabling infrastructure and act accordingly. For this reason, we broadly support the Federal Government's release of *Expectations of data centres and AI infrastructure developers*. Data centres are already meeting or exceeding these expectations. The NSW Government's release of a public consultation paper on competing principles works to muddy the waters for investment in Australia, and should carefully consider how it aligns or not with the national expectations.

The central policy question is clear. How can Australia capture the economic value created by data centres while managing the infrastructure systems that support them? The answer begins with recognising what data centres really are: the foundation of the services Australians rely on every day.

2. Key recommendations

Recommendation 1.

Recognise data centres as a strategic economic opportunity for NSW and recalibrate policy settings, regulatory approaches and community narratives to enable and attract investment at scale

Recommendation 2.

The NSW approach to data centres should align with the Commonwealth Government's *Expectations of data centres and AI infrastructure developers* to create a national and consistent approach so Australia can compete for investment and streamline national data centre operations.

Recommendation 3.

The NSW Government should give the new Investment Delivery Authority the ability to rapidly assess and recommend public infrastructure investments that support major private sector projects.

Recommendation 4.

Acknowledging recent positive reforms to NSW's planning system, the NSW Government should monitor reforms to ensure they deliver better performance, focusing on efficiency, consistency, and certainty in the planning system.

3. The human benefit: why data centres are essential

3.1 Supporting everything we do

A normal day in modern Australia already involves constant interaction with data centres, even if most people never notice it. Sending a message, transferring money, taking public transport, booking a medical appointment, joining a video call or streaming a show all rely on data centres operating in the background.

For many people, the concept of a data centre is unfamiliar, and this lack of understanding can distort how they are understood. The public debate can sometimes reduce them to symbols of trivial internet content. In reality, they support the routine functioning of many systems people rely on every day.

Case Study: Amy's homework

Amy lives in Parramatta and is a Year 10 student. One night she uploads her maths assignment before the deadline.

When Amy presses "submit," the file travels through the internet to a data centre. Inside the building, powerful computers store the document safely in the cloud. Her teacher can open the assignment from school the next morning.

Data centres quietly store schoolwork, photos, messages and videos for millions of Australians every day, making it possible to access files from almost anywhere.

3.2 A digitised world

Hospitals, banks, government services, transport networks and communication platforms all rely on data infrastructure to function reliably and securely. These services existed long before modern data centres, but they operated on a far smaller scale. In the past, these processes relied on paper records and local computers that handled limited amounts of information.

Over time, the scale and expectations of these systems have changed dramatically. Banks now process millions of digital transactions every day through online and mobile platforms. Hospitals store and access large volumes of electronic health records, diagnostic images and laboratory results. These services must operate at all hours, with high levels of reliability, security and speed. Data centres provide the infrastructure to store and process this information at scale, keep systems available around the clock and protect sensitive financial and medical data, amongst many other types of information and processes. As society has digitised, data centres have shifted from being optional support systems to essential infrastructure.

Data centres strengthen the wider economy rather than act as a burden on it. They convert electricity and computing capacity into productivity across every sector. Businesses operate more efficiently, services run more smoothly and new digital industries can grow. Yet community trust in their expansion depends on demonstrating clearly that the infrastructure serves people first and delivers broad economic value.

Case Study: Margaret's health monitor

Margaret is 78 and lives alone in Sydney. She has a heart condition, so her doctor asks her to wear a small health monitor on her wrist. The device tracks her heart rate and blood pressure throughout the day. Every few minutes it sends this information through the internet to a secure data centre.

Inside the data centre, servers store Margaret's health data and run software that looks for unusual patterns, such as a sudden change in heart rhythm. If the system detects a risk, it alerts her doctor or nurse.

This allows medical teams to monitor elderly patients remotely, helping them stay independent at home while still receiving continuous medical oversight.

Recommendation 1.

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3.3 Community trust and public confidence

Trust and confidence in data centre development depends on whether communities see these facilities as serving a clear public purpose. Debates about new infrastructure are often framed around environmental impact, and they also concern whether infrastructure contributes to social wellbeing.

Data centres support services that many Australians consider essential and help sustain a more inclusive digital society. Remote communities can access services that were once limited to major cities. Patients can receive telehealth consultations. Students can learn through online platforms. Small businesses can reach customers and run their operations using affordable cloud tools rather than expensive local IT systems. Digital infrastructure reduces barriers to participation in the economy and public services.

Community concerns about environmental impacts are legitimate and deserve attention. Data centres use energy and water, but they are also the most efficient ways to deliver computing services at scale. By centralising digital workloads in specialised facilities, they use far less energy than the equivalent thousands of smaller on-site servers and devices scattered across workplaces and homes.

For many voters concerned about climate, community and fairness, the key question is how data centres are built and governed. Clear reporting from government on energy and water use, responsible infrastructure planning and visible local benefits will build confidence in the sector. Better, more reliable information about the benefits and impacts of data centres will help dispel some prevalent myths about them. When development is transparent and impacts managed responsibly, data centres become part of the infrastructure that supports a cleaner, more connected and more productive society.

3.4 Different types of data centres

Data centres are not all the same.

Very simply, there are three main types of data centre:

- **Cloud** – Most large facilities in Australia are cloud data centres. These sites host cloud platforms that provide computing, storage and software services to businesses, governments and households over the internet. Cloud workloads often favour large campuses in major cities such as Sydney and Melbourne, where network connections and customers are concentrated.
- **AI training** – These facilities support AI workloads, which require more specialised computing hardware such as graphics processing units (GPUs). In places like the US and China, training data centres run massive computing jobs that build and refine general purpose AI models, but Australia currently does not host this type of data centre. ‘AI training’ done in Australia is of a much smaller scale, for narrow purposes, such as scam detection in banks² and computer-vision systems for mining safety³, and is more accurately described as either machine learning models or fine tuning of pre-trained models.

² Westpac, *Using AI to put scammers out of business*, 29 May 2025

³ BHP, *AI is improving performance across global mining operations*, 30 January 2026

- **AI inference** – This function mostly sits within the cloud data centres described above. It is articulated here to highlight the differences between AI training and inferences. AI inference means running trained models in everyday applications such as chatbots, fraud detection or recommendation systems. Inference workloads run continuously and are often located closer to users to reduce latency.

These different types of data centres form the compute backbone for cloud services, digital platforms and AI capabilities across Australia's economy.

Another way of categorising data centres is by size, defined by the maximum electrical power capacity available to the data centre. There is no hard and fast definition of data centre types and sizes, and the sector is evolving quickly.

Type	Description	Size
Hyperscaler	Facilities built and run by global tech giants including AWS, Google, and Microsoft. These sites operate at very large scale and provide the infrastructure behind cloud platforms, online applications and AI workloads.	50–300 MW +
Co-location and service providers	Operators lease either space for customers to house their own equipment (co-location) or provide space and equipment (service providers). However, sometimes this can also mean providing hyperscale services for hyperscale customers.	25–200 MW +
Enterprise / on-site	Run by organisations for their own use, often on their own premises. Banks and telcos often run data centres around 20 MW. Large university research clusters and government data centres can reach 100 MW.	10–100 MW
Edge	Low-latency processing for local end users.	<10 MW

Table 1: Data centre types by size

3.5 Uplifting the entire economy

Beyond their role as essential infrastructure, data centres generate unusually high economic value relative to the resources they consume. Digital services delivered through data centres support productivity across the entire economy.

One way to understand this contribution is to compare how much economic value different sectors produce from the energy they use. The technology sector generates about \$12.60 in gross value added for every kilowatt hour of energy consumed (about the same as running a household air conditioner for an hour), higher than other major sectors.⁴ The technology sector contributed around \$250 billion to the economy in 2025, or about 9 per cent of GDP.⁵

The point is not that one sector is more important than another. The key difference is that digital infrastructure enables productivity across all sectors at once. Data centres support the software platforms used by miners and the logistics systems used by manufacturers and the online services used by consumers. In this way, they act as a multiplier for the broader economy rather than a stand-alone industry.

Case Study: AWS cloud for Commonwealth Bank⁶

Commonwealth Bank has moved its banking system into cloud-based data centres operated by Amazon Web Services. This includes customer ledgers, transaction processing, and core banking functions. It is one of the largest cloud migrations undertaken by an Australian bank.

⁴ Mandala, *Data Centres as Enabling Infrastructure*, November 2025, p.1

⁵ Tech Council of Australia, *Technology as Australia's Productivity Engine*, 2026, p.1

⁶ James Eysers, *CBA moves its entire core banking system into Amazon's cloud*, Australian Financial Review, 6 October 2025

Data centres determine how quickly financial services can evolve, scale, and integrate artificial intelligence into real-time operations. The bank expects costs to be similar to its previous in-house systems. The gain is speed. Product changes that once took months can now be deployed in weeks. Customer-facing performance has improved, with faster response times across digital channels.

This case shows how data centres function as critical infrastructure. In this case, AWS's data centres now process around two in five transactions in the Australian economy.

3.6 Australia's future in the AI value chain

AI is turning intelligence into infrastructure. Value is shifting away from individual knowledge-based work and towards intelligence generated by access to compute, data centres, and energy.

Control of that infrastructure now matters. Nations that build and house that infrastructure will see more of the benefits of AI.

The upside is large. Countries that invest early will see massive productivity gains across finance, healthcare, logistics, education and government. These gains build over time and flow through the whole economy. The downside is just as clear. Countries that do not build this capability will rely on offshore platforms. That creates long-term dependence and limits strategic choice.

Australia starts from a position of strength and global demand is moving in Australia's direction. International data centre operators are selecting Australia for new capacity, drawn by local demand, government stability, connectivity, access to energy and Australia's favourable positions in the five eyes alliance and other multilateral frameworks.

3.7 How data centres are lifting investment in Australia

Australia hosts a large and growing group of data centre operators and infrastructure investors. Companies such as AirTrunk, Telstra, AWS, Microsoft, Google, and Goodman who build and operate facilities across the country. These companies employ thousands of Australians, pay local taxes and reinvest heavily in new infrastructure.

Hyperscale providers do not just build and operate their own facilities. They are also the anchor customers for a large share of Australia's data centre capacity through long-term leasing arrangements. That means much of the revenue and enterprise value captured by Australian home-grown data centre operators is tied directly to contracts to provide services to companies like Microsoft, AWS and Google.

When global cloud providers invest in Australia, much of the economic activity remains local. Data centres require large construction projects, electrical infrastructure, fibre networks and ongoing operational staff. Australian engineering firms, electricians, construction workers and equipment suppliers, among others, all participate in building and maintaining these facilities.

The benefits also extend beyond the data centre sector itself. Australian businesses, governments and education providers use cloud platforms hosted in local facilities. Australian start-ups can launch services without building their own IT infrastructure, while established companies can adopt advanced analytics and AI tools that improve productivity.

Seen in this light, data centres do not simply generate profits for offshore technology firms. Not only do data centre operators in Australia pay full taxes with no concessions, they form the infrastructure that allows Australian companies, workers and communities to participate fully in the digital economy. Their growth in response to local demand strengthens national capability, adds to the tax base, creates high-skill jobs and supports productivity across the entire economic system. Data centre growth is an indicator that the digital economy is growing.

3.8 NSW as part of a national approach

A strong data centre sector in NSW contributes to the broader national interest.

Compute infrastructure now underpins national capability in AI and digital services. Hosting computing capacity within Australia strengthens resilience across critical sectors such as finance, defence, healthcare and public administration. When data and computing systems operate domestically, governments and organisations maintain greater legal control, stronger security oversight and more reliable access to critical digital services.

Australia holds several structural advantages that position us well to host this infrastructure.

“

Our abundant renewable energy potential, robust privacy protections and strategic Indo-Pacific location can make Australia an AI hub for the region.

Australian Government, National AI Plan, December 2025

”

Geopolitical tensions around technology supply chains have further elevated the importance of domestic computing infrastructure. In this environment, data centres are no longer only commercial assets. They form part of the strategic infrastructure that supports national resilience.

NSW state policy should operate within a coherent national framework. The Commonwealth Government has released national expectations for data centre development.⁷ Alignment between federal and state settings will be essential. A patchwork of inconsistent rules and expectations across jurisdictions would add complexity and discourage investment in Australian infrastructure, reducing our overall national resilience. As NSW starts its consultation on its own data centre principles, the most critical consideration will be harmonisation with the national expectations. Australia needs to reduce complexity, not potentially increase it with unaligned and competing expectations across jurisdictions.

At the same time, any policy settings, expectations or guidelines should avoid becoming overly technical or prescriptive. Rapid advances in technology mean that rigid rules quickly become outdated. Data centre operators – along with the rest of the economy – are already facing a growing compliance burden, making us less competitive.⁸ A clear set of national expectations, applied consistently across jurisdictions, provides certainty for investors while allowing flexibility as technology and energy systems continue to evolve.

Recommendation 2.

The NSW approach to data centres should align with the Commonwealth Government's *Expectations of data centres and AI infrastructure developers* to create a national and consistent approach so Australia can compete for investment and streamline national data centre operations.

⁷ Department of Industry, Science and Resources, *Expectations of data centres and AI infrastructure developers*, 23 March 2026

⁸ Business Council of Australia, *BCA Global Investment Competitiveness Index*, February 2026

4. An energy growth partnership

The energy requirements of a large and growing data centre sector present both challenges and opportunities. The sector's scale means it can serve as a genuine partner in the state's renewable energy transition. Data centres can help anchor new renewable energy investment and support the expansion of modern grid transmission infrastructure.

AEMO's estimates suggest that data centres consumed around two per cent of grid-supplied electricity in 2024–25, with demand expected to rise to around six per cent by 2029–30.⁹ AEMO's figures are the most reliable as they come from metered electricity use. Data centre operators must report forecasts to AEMO under the National Electricity Rules, so it holds the most accurate view of actual and expected demand.

These figures highlight the importance of planning rather than alarm, and the need for accurate information rather than speculative estimates. Growth in electricity demand is a normal feature of economic development and technological change. When coordinated with renewable generation, energy storage and network investment, the growth of data centres can support the broader transition of Australia's energy system while strengthening the infrastructure that underpins our digital economy.

4.1 A net reduction in energy burden

One of the most counterintuitive things about data centres is that they *reduce* rather than increase the total energy consumed by computing. Data centres are an efficiency creating mechanism.

The alternative to modern data centres is a dispersed system in which computing takes place across thousands of small, on-premise servers and devices located in homes and office buildings. These systems typically run at low utilisation and use less efficient hardware. As a result, they consume far more electricity overall. If all data centres suddenly disappeared, we would still need computers to do the same work, but we would require seven times more electricity to do it.¹⁰

Clearly, data centres are the most efficient way to manage our digital society in a sustainable way. They aggregate computing workloads in purpose-built environments where operators can manage power use carefully, maintain high equipment utilisation and deploy advanced cooling systems. Modern facilities use sophisticated power management systems and highly efficient chips specialised for different types of workloads that maximise the amount of computing delivered per unit of electricity.

4.2 Increasing efficiency

Data centres are only getting more efficient. Efficiency gains can be measured through Power Usage Effectiveness, or PUE, the standard global metric for data centre efficiency. In 2007 the global average PUE was around 2.5, meaning that for every unit of electricity used for computing, another 1.5 units were consumed by cooling and other support systems. Today the global average has fallen to around 1.58.¹¹ Australian facilities perform even better. Many report design PUE levels as low as 1.15, with median values around 1.30.¹²

Specific hardware is also getting more efficient. Modern AI chips have historically doubled their energy efficiency roughly every two years.¹³ This means that over time, many data centres will perform far more computing work while using comparatively less electricity.

⁹ Australian Energy Market Operator, *2025 Inputs, Assumptions and Scenarios Report*, August 2025, p.110

¹⁰ Mandala, *Data Centres as Enabling Infrastructure*, November 2025 p.18

¹¹ Daniel Bizo, *Global PUEs — are they going anywhere?*, Uptime Institute, 4 December 2023

¹² Mandala, *Empowering Australia's Digital Future*, October 2024, p.6

¹³ Robi Rahman, *Leading ML hardware becomes 40% more energy-efficient each year*, Epoch AI, 23 October 2024

4.3 Building in Australia is better for global emissions

The global demand for computing is growing rapidly, particularly as AI and digital services expand. The infrastructure required to support this demand will be built somewhere.

Australia has reasons beyond economics for wanting to be that ‘somewhere’. Australia’s world-leading renewable energy commitments and expanding clean energy supply mean that hosting data centres here will reduce the carbon intensity of global computing. If these facilities are not built in Australia, they are likely to be built in countries where electricity systems remain far more emissions intensive.

4.4 Anchoring the renewable transition

The economics of renewable energy projects depend heavily on the certainty of long-term revenue. Building a large wind or solar project requires substantial upfront investment, and developers typically cannot secure project finance without clear expectations about who will purchase the electricity (at what price) once it is generated.

Data centre operators are well placed to provide this certainty. Their facilities create stable and predictable demand. This makes them attractive long-term customers for renewable energy developers. When data centre operators commit to long-term electricity contracts, they underpin the financing of new wind and solar capacity that might otherwise struggle to proceed.¹⁴ The growth of data centres helps accelerate investment in renewable generation and storage, supporting a more resilient and lower-emissions electricity system.

Data centre operators are already contributing to this transition through investment in grid infrastructure. Since 2020, companies such as Amazon, Microsoft and Equinix have contracted close to one gigawatt of renewable electricity capacity from wind and solar projects in Australia.¹⁵

Across the sector, around 70 per cent of electricity used by data centres is already offset by renewable energy, through a mix of on-site solar generation and long-term renewable power purchase agreements (PPAs) with renewable energy developers to secure electricity supply.¹⁶

Case Study: Microsoft and Walla Walla Solar Farm¹⁷

The Clean Energy Finance Corporation (CEFC) initially committed \$100 million, alongside ING and Export Development Canada, to support a 300 MW solar farm in Walla Walla, NSW. In July 2024, this shifted to a broader CEFC commitment backing a 1 GW portfolio of FRV Australia solar and battery projects, including Walla Walla.

FRV Australia is a major solar developer with a long track record in the local market. Walla Walla will deploy around 700,000 single-axis tracking panels and generate enough electricity to power about 90,000 NSW homes. It connects into existing transmission in a strong part of the grid and supports continued sheep grazing.

The project is anchored by Microsoft. A 15-year power purchase agreement provides long-term revenue certainty and directly supports Microsoft’s Australian data centre operations. This underpins Microsoft’s commitment to run on 100 per cent renewable energy and its target to be carbon negative by 2030, with full historical emissions removal by 2050.

¹⁴ Baringa and CEFC, *Getting the balance right: data centre growth and the energy transition*, December 2025, p.3

¹⁵ Ryan Cropp and Amelia McGuire, *Tech giants pitch data centres as climate saviours – not threats*, Australian Financial Review, 25 September 2025

¹⁶ Mandala, *Data Centres as Enabling Infrastructure*, November 2025, p.16

¹⁷ CEFC, *Powering Microsoft data centres with Walla Walla sunshine*, August 2024

4.5 Phantom demand

The rapid growth in data centre investment has produced a surge in applications for electricity grid connections. Yet not every application represents a project that will be built. Many prospective developers lodge connection requests early in the planning process to secure potential access to network capacity, even if the project itself remains uncertain.

This dynamic has created what energy planners call “phantom demand.” In 2025 the Australian Energy Market Operator (AEMO) received around 44 gigawatts of connection requests linked to data centre proposals.¹⁸ Only about six gigawatts of project capacity is expected to proceed, with only 2.8 gigawatts of actual electricity draw once those facilities reach maturity – 15 times lower than what is sometimes reported.

Energy planners are already aware of this issue. AEMO has incorporated these dynamics into its modelling by filtering connection requests and focusing on projects that show credible evidence of development.¹⁹

Even so, phantom demand can shape public debate. Large numbers attached to connection requests creates the misperception that electricity demand from data centres will be far higher than the actual demand. Understanding the difference between connection requests and actual electricity use is therefore critical for informed policy discussion.

4.6 Connections and transmission

Data centres in Australia pay for their own connections and any network upgrades – like substations – that are needed, unlike much of the US.

The central infrastructure challenge in Australia is not simply electricity generation. Australia has substantial renewable energy potential. The more immediate constraints lie in electricity storage and transmission, which are necessary to move power from where it is generated to where it is needed and to balance variable renewable supply.

National energy planning suggests that around 6,000 kilometres of additional transmission lines may be required by 2050, representing roughly a 13 per cent expansion of the network.²⁰ Expanding storage and transmission capacity will be essential to support both renewable generation and new electricity demand from emerging industries. AEMO highlights other major drivers of demand, including electrification of industry and households, along with rapid growth in electric vehicles. The point is that the grid is shared infrastructure for the whole economy, not just data centres.

¹⁸ Oxford Economics, *Surging Data Centre Connection Requests Drive Phantom Demand*, 21 November 2025, pp.1–2

¹⁹ Oxford Economics, *Surging Data Centre Connection Requests Drive Phantom Demand*, 21 November 2025, pp.1–2

²⁰ Australian Energy Market Operator, 2025 Information Toolkit - Integrated System Plan, p.21

5. Managing water

Water is among the most sensitive resource questions in a state that experiences periodic drought, faces long-term pressures from population growth, and is acutely aware of the need to manage natural resources carefully.

5.1 Proportionate potable water use

Current water use by data centres is small relative to overall supply. Similar to energy use, when use of water is aggregated into data centres, it becomes far more efficient. Across Australia, data centres consume around 5.5 gigalitres of water each year, which represents approximately 0.04 per cent of national potable water use. In Sydney, data centres account for about 0.7 per cent of total water consumption.²¹ Even with continued expansion of the sector, industry projections indicate that data centre water use is expected to remain below two per cent of total potable water supply through to 2030.

5.2 Using recycled water

Data centre operators are investing in recycled water infrastructure, with projected industry investment estimated to reach between \$500 million and \$1.1 billion by 2030.²²

Recycled water infrastructure benefits the broader community by reducing reliance on potable supplies and improving the overall efficiency of the urban water cycle. Data centres, which can use non-potable water for cooling systems, have a strong commercial incentive to support these investments. For example, recycled water now supplies 55 per cent of AirTrunk's total use, and its Singapore site runs entirely on reclaimed water.²³

When recycled water networks are available, operators reduce operating costs while easing pressure on drinking water supplies.

Government planning frameworks play an important role in enabling these investments. Utility partnerships, clear planning approvals and infrastructure coordination make recycled water projects more viable. Data centres have shown a clear willingness to use recycled water and fund the infrastructure needed to connect to supply. But in NSW, access remains constrained. Policy settings should support broader availability of recycled water for industrial users, including data centres, where demand and funding are already in place.

5.3 Better cooling technology

Cooling technology continues to evolve in ways that reduce water demand.²⁴ Modern data centres can operate cooling systems which significantly reduce or eliminate the need for continuous water supply.

Closed-loop liquid cooling uses far less water than most people expect. The system circulates a fixed volume of coolant through sealed pipes. That liquid absorbs heat from chips, then releases it through a heat exchanger. The same fluid cycles again and again. It is not evaporated, and it is not discharged. Water only enters the system in small amounts for initial filling or occasional top-ups. In steady operation, consumption is close to zero.

Air cooled systems can also reduce water consumption, but they do it in a different way. Many modern data centres use dry cooling, where heat is rejected to ambient air through radiators or cooling coils. These systems rely on fans and heat exchangers, not evaporation. As a result, they use little or no water during normal operation. Dry cooling works well in milder conditions, but its efficiency drops during hot summer periods, so systems switch to evaporative cooling, which uses water but reduces electricity demand, reflecting a trade-off between water use and energy efficiency.

²¹ Australian Bureau of Statistics, *Water Account*, Australia, Table 1.1., 2023

²² Mandala, *Data Centres as Enabling Infrastructure*, 2025, p.1

²³ AirTrunk, *Sustainability Report FY25*, 2025, p.7

²⁴ Mandala, *Empowering Australia's Digital Future*, p.28, 2024

These developments show that water use in data centres is manageable, increasingly flexible and the means of minimising electricity use. With the right combination of recycled water infrastructure, transparent reporting and modern cooling technologies, the sector can expand while limiting pressure on water resources and ensuring that the digital economy grows in a sustainable way.

6. Increasing connectivity

As digital services scale, customers increasingly depend on low latency, resilient, and predictable connectivity between applications, users, cloud platforms, and partners. Long-distance transit paths often introduce latency, congestion, and cost inefficiencies, particularly for cloud, AI, media, and real time applications. What this means is that where digital infrastructure is located and the quality of that digital infrastructure matters greatly.

Modern data centres are no longer just places to house servers. They frequently become connectivity hubs attracting carriers, cloud providers, content delivery networks, and enterprises. This provides opportunities for customers and local residents to gain access to multiple carriers, paths, and platforms within the same location, improving redundancy and reducing dependency on any single network or route.

6.1 AI increasing network demand

Data centres already move a lot of data around. AI is increasing the amount of data that digital networks must handle. AI systems rely on extremely large datasets, powerful computing infrastructure and the ability to move information quickly between different locations.

Fast and reliable connections between data centres, cloud platforms and end users are essential for applications such as real-time analytics and AI inference systems that respond instantly to new information. This needs high-speed, resilient connectivity to support AI-ready data centres and cloud platforms, enabling Australia to process, move and use data securely, reliably and at scale.

Fully realising AI's potential requires investments in strengthening connectivity, resilience and security across the data and AI ecosystem, and in building foundations that are fit for purpose as AI adoption accelerates. Without modern data and connectivity foundations, AI initiatives will struggle to scale safely or deliver sustained value.

Case Study: Telstra's Aura Network build

Telstra is building a new next generation high-capacity low-latency backhaul fibre network spanning almost 14,000 kilometres across Australia. The dual-architecture design provides an express network between capital cities, and a foundation network that enables future expansion into regional and remote areas across the country. While this national fibre infrastructure is critical to supporting digital economies and data centre operations, the Aura Network is a complex project, requiring construction across diverse environments and intensive coordination with hundreds of landowners and land managers, highlighting opportunities to improve delivery efficiencies.

On the Sydney–Canberra–Melbourne coastal route alone, more than 6,000 separate regulatory and site access actions have been required to deliver just over 1,000 kilometres of fibre. That equates to roughly six separate approvals, notices or assessments for every kilometre of network built.

This highlights the administrative complexity involved in deploying national digital infrastructure, even for a single fibre corridor between major cities.²⁵

6.2 Edge computing, fibre networks and subsea cables

Edge computing is becoming an important part of this architecture. Edge systems process information closer to where it is created, lowering latency and easing congestion on core networks. Physically, edge data centres are much smaller than traditional facilities. Many have a capacity of two megawatts or less and can occupy a modular container or a few server racks within an existing structure.

Investment in fibre networks and subsea cables is also important to expand domestic and global connectivity. Metro fibre networks are being strengthened to support high-capacity traffic within major cities. New subsea

²⁵ Vicki Brady, *A shared national vision for Australia's digital future*, Telstra, 10 September 2025

cables are expanding the country's international connectivity. Data centre operators like Google²⁶ and Meta²⁷ are making heavy investments into subsea cable connectivity globally.

These investments highlight that connectivity infrastructure forms part of a nation's strategic digital capability. Reliable and secure connectivity strengthens economic resilience, supports national security and allows Australia to participate fully in the global digital economy.

Case Study: Edge computing²⁸

An edge data centre in Dubbo NSW, opened in 2021, shows how regional infrastructure can support national digital growth. The facility places compute capacity closer to users in western New South Wales. This reduces latency and improves performance for services such as healthcare, education, and local business systems.

The project brings direct economic benefits. Construction creates local jobs, and ongoing operations require skilled technicians and support staff. It also attracts new digital businesses that need reliable, low-latency services outside capital cities.

Energy and land availability in regional areas support this model. Dubbo offers lower costs and room to expand, which is harder to secure in Sydney. At the same time, the data centre strengthens local resilience by keeping critical data and services closer to communities.

This case shows how edge data centres extend national infrastructure into regional Australia. They improve service quality, support local economies, and reduce pressure on major city hubs.

²⁶ Brian Quigley, *Connecting the South Pacific with new subsea cables*, Google Cloud, 26 October 2026

²⁷ Emily Forlini, *Inside Meta's Effort to Build the World's Longest Underwater Cable*, PC Mag, 4 December 2025

²⁸ Leading Edge DC, *Leading Edge Opens Dubbo Data Centre, Forecasting 621 New Jobs and \$106m per Year in Economic Growth For the Region*, 13 December 2021

7. Place and community

Data centres are often large infrastructure projects with visible impacts on the communities where they are located. They require land, electricity connections, fibre networks, transport access and access to a skilled workforce to keep them operating. Planning decisions about data centre development must recognise these local impacts. They should also recognise the benefits these facilities can deliver to host communities and to the wider NSW economy.

7.1 Supporting private infrastructure delivery

Private infrastructure projects often require supporting public infrastructure. A new data centre, warehouse, or logistics hub needs reliable power, telecommunications, water services, and road access. These systems allow the facility to operate and connect with the wider economy. These projects create jobs, expand economic activity, and generate tax revenue.

The NSW Government already recognises the link between private investment and public infrastructure. Many jurisdictions have created teams that help coordinate approvals and guide investors through planning and regulatory processes. These teams need a clear understanding of barriers so they can support delivery of digital infrastructure.

Business groups report that governments could strengthen this support. One issue is the timing of local infrastructure investment. Private investors often identify sites before supporting infrastructure arrives. A company may choose a site for a data centre. The project may require electricity upgrades, water connections, or local road improvements. These works may appear in long term government plans but remain years away from delivery.

Bringing these upgrades forward can unlock immediate private investment. The project proceeds sooner and the surrounding precinct becomes more attractive for other businesses.

The NSW Government should take the Investment Delivery Authority's coordination role further, creating a process that allows credible investors to trigger rapid assessment of these upgrades. The process would not guarantee public funding, but would provide a pathway to feed into agency and government processes, to quickly assess whether earlier infrastructure delivery provides clear economic benefits. This may require adjusting Objective 2 - Criteria 2 (reliance on government funding) of the Investment Delivery Authority EOI process, to recognise that supporting government infrastructure investment may flow from the decision.

Recommendation 3.

The NSW Government should give the new Investment Delivery Authority the ability to rapidly assess and recommend public infrastructure investments that support major private sector projects.²⁹

7.2 Approval speed

Planning and approval processes must also move more quickly to keep pace with global investment cycles. Data centre investment is highly mobile. Many developers can choose between multiple international locations, and long approval timelines can shift projects to other jurisdictions.

Responsibility for planning approvals largely sits with states, territories and local government. Clearer timelines, efficient assessment processes and coordinated agency decision making would help improve certainty for investors.

²⁹ Business Council of Australia, *Building Australia's Tomorrow*, 2025, p.61

The objective should be to streamline and facilitate development, recognising the economic value data centres generate and the broader uplift they deliver. The BCA has consistently called for a reduction in regulatory burden, and we do not support introducing additional regulation where it is unnecessary.

The NSW Government is well aware it has room to improve in this area, and the BCA supports many of the planning reforms the Government is pursuing. In the BCA's 2025 *Regulation Rumble*, the state ranked last as at end June 2025 for planning system performance compared to other jurisdictions across the country.³⁰ The state performs well in planning process transparency, but efficiency, consistency, and certainty for proponents lagged behind most other states and territories.

Addressing this gap will be important if the state wishes to remain competitive for major digital infrastructure investment, and initiatives like updated planning laws, the Development Coordination Authority, and Investment NSW are all positive recent initiatives. Ultimately, the success of these will be determined by their real-world impacts over the coming months and years. The NSW Government should not assume that these initiatives are all that is required – it needs to continue to improve planning system performance.

Recommendation 4.

Acknowledging recent positive reforms to NSW's planning system, the NSW Government should monitor reforms to ensure they deliver better performance, focusing on efficiency, consistency, and certainty in the planning system.

7.3 High-value local supply chains and employment

NSW is entering a period of rapid expansion in data centre and AI infrastructure. The key policy question is how the state can capture lasting economic value from this growth.

The answer lies in building a strong local ecosystem around the sector. This includes deeper supply chains, a larger skilled workforce and the development of complementary industries.

Data centre development relies on a wide range of specialised capabilities. Facilities require advanced construction skills, electrical and mechanical engineering services, cooling and control systems expertise, fibre and network infrastructure, physical and cyber security, and regulatory compliance services. They support cloud solution architects, software programmers, local software companies, AI developers and digital service providers who are working in the sector already.

If the state can work with data centre developers to facilitate predictable pipelines of projects and clear workforce training pathways, businesses will invest in building these capabilities locally. This creates a reinforcing cycle in which infrastructure development supports the growth of skilled jobs and specialised industries.

Workforce planning will be particularly important. Estimates suggest the Australian data centre workforce will need to grow by around 8,300 people, reaching approximately 17,900 workers by 2030.³¹ States play a central role in championing the value of careers in data centre development and maintenance, and in funding vocational training, apprenticeships and technical education. Expanding these pathways will help ensure that local workers benefit from the growth of the digital infrastructure sector while supporting the skills base required for future investment.

³⁰ Business Council of Australia, *Regulation Rumble 2025*

³¹ Mandala, *Empowering Australia's Digital Future*, October 2024, p.6

8. Urgency of the opportunity

Global competition for data centre investment is intense and growing. Governments across the Asia-Pacific region recognise that digital infrastructure will underpin future economic growth, AI capability and technological development. Capital is moving quickly into this sector, and jurisdictions are competing to attract projects that bring computing capacity, high-value jobs and supporting industries.

Australia, and NSW in particular, currently holds a strong position. The state offers political stability, a reliable rule of law, a high-quality fibre network and proximity to the rapidly expanding economies of the Asia-Pacific region. These advantages make it an attractive location for digital infrastructure investment. Yet they should not be assumed to guarantee future projects.

Demand for data centre capacity remains extremely strong. Globally, AI-focused data centres operate at around 98 per cent utilisation, which indicates that the risk of overbuilding in the near term is very low.³² Data centre vacancy rates in Australia also remain tight. In Sydney, the vacancy rate in the second quarter of 2025 was about six per cent, reflecting strong demand for existing facilities.³³

The project pipeline in NSW illustrates the scale of the opportunity. Around \$30 billion of data centre investment, representing roughly three gigawatts of capacity, is currently moving through different stages of planning and approval.³⁴

But developers and investors have choices. Singapore, Japan, Malaysia and several other Asia-Pacific markets are actively competing for the same capital. Many offer streamlined approval processes, dedicated electricity infrastructure and explicit government partnerships with industry.

If planning systems are slow, unpredictable or impose costs that competing jurisdictions do not, capital will move elsewhere. Digital infrastructure investment tends to follow certainty and speed. The case for a clear and supportive data centre policy in NSW is ultimately a case for productivity and living standards.

³² CBRE Research, *North America Data Center Trends H1 2025: AI & Hyperscaler Demand Lead to Record-Low Vacancy*, 2025

³³ Colliers Research, *The Race for Data Sovereignty: Why More Australian Data is Staying Home*, September 2025, p.12

³⁴ Commonwealth Bank of Australia, *Renewables and Data Centres: Australia's next mining boom?*, December 2025, p.4

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